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Jeffrey B. Halverson & Thomas Rabenhorst

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HURRICANE SANDY:

The Science and Impacts of a Superstorm

by Jeffrey B. Halverson and
Thomas Rabenhorst

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Manhattan suffered a widespread power outage during the storm.



It was the storm of a lifetime—a massive, **freakish confluence of a tropical hurricane and a winter, extratropical vortex**. Nicknamed “Frankenstorm,” Superstorm Sandy ravaged the mid-Atlantic, Northeast, and Ohio Valley regions for three days from October 29-31, 2012. Hollywood could not have scripted a more bizarre landscape of post-storm devastation: **Lower Manhattan lay submerged and in the dark**. Many of New Jersey’s beaches shrank widthwise by 30-40 feet. Numerous fires ignited from ruptured natural gas mains. Power outages approaching 10 million enveloped a massive region from Virginia to Maine to Michigan. Mile-long lines plagued gas stations, and fuel was being rationed according to odd/even days. Jumbled heaps of debris, sodden with mud and mold, were all that remained of homes in dozens of coastal towns.

This was Superstorm Sandy. The 2012 Atlantic hurricane season was once again in hyperactive mode, breeding 19 named storms. Since 1995, most years have tallied storm counts far exceeding the long-term average of 11 named systems. And in recent years hurricanes have become exceptionally large, including Isabel (2003), Ike (2008), and Irene (2011). But something as outlandish as Sandy seemed to strain the limits of Mother Nature.

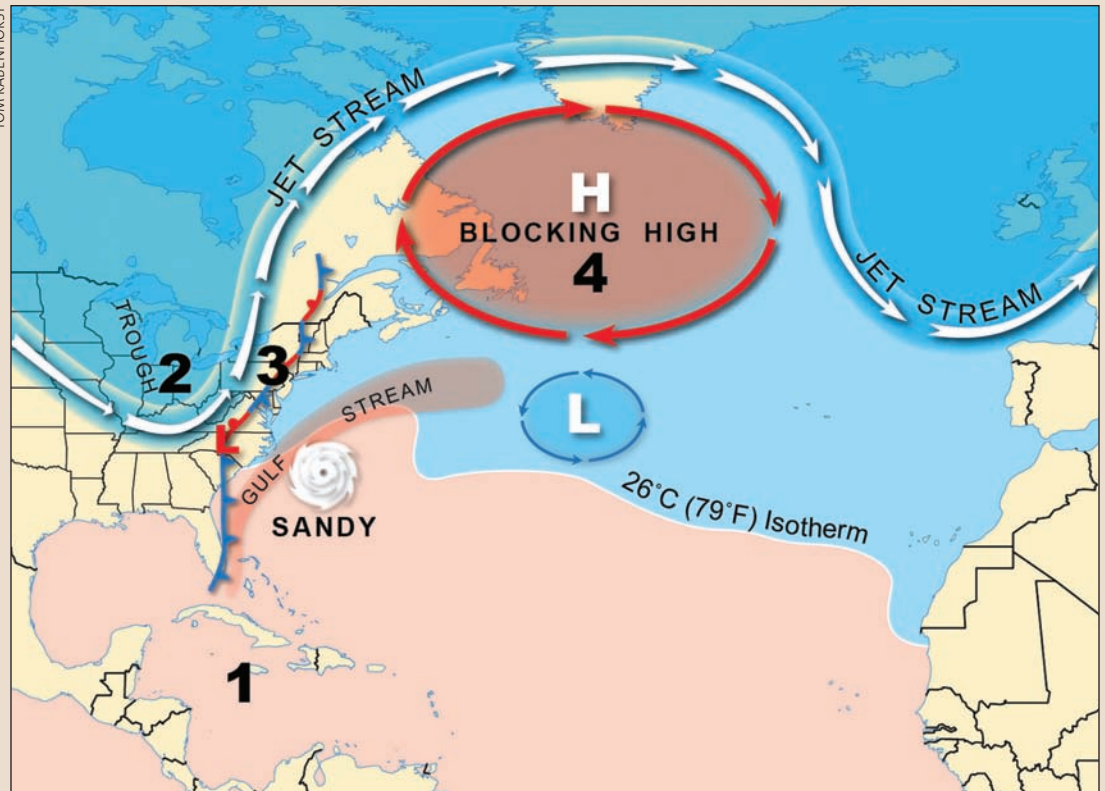


Figure 1. A visual roadmap of the major atmospheric and oceanic elements that conspired to produce the superstorm.

Figure 1 provides a visual roadmap of the major atmospheric and oceanic elements that conspired to produce the superstorm. The various pieces were highly dispersed, from the Caribbean to Greenland to the Great Lakes. The storm began as an unusually intense, late-season hurricane south of Cuba, which rapidly intensified to nearly Category 3 hurricane status. This required very warm ocean surface waters—Element 1 in Figure 1. Sandy's passage across the mountainous terrain of Cuba temporarily knocked the wind from its sails. But after weakening to a humble Category 1, Sandy began to grow in size. And continue to grow. The maelstrom eventually swelled to more than 1,100 statute miles in diameter, becoming the largest tropical cyclone in Atlantic basin history. Two days before landfall, Sandy developed a system of weather fronts. It bore resemblance to a large, winter-like Nor'easter, yet with a distinctly tropical core. The only word to describe such a cyclone is a "hybrid." A large, intense trough in the mid-latitude jet stream (Figure 1, Element 2) and its surface cold front (Figure 1, Element 3) contributed directly to the transition of Hurricane Sandy to a hybrid storm. And Sandy was the only tropical cyclone in United States history for which both hurricane and blizzard warnings were issued simultaneously.

Even more amazing, a peculiar configuration of atmospheric steering currents thrust the burgeoning vortex westward, against the grain of the

mid-latitude's prevailing circulation. It was the worst possible scenario for a coastline teeming with tens of millions of inhabitants. Element 2 in Figure 1, the jet stream trough, and Element 4 in Figure 1, a blocking ridge of high pressure over Greenland, interacted to turn Sandy toward the coast. A great many meteorologists were at a loss to explain the improbability of this combination of events.

In this article, we explore the meteorological life story of Superstorm Sandy. One of the more interesting attributes was Sandy's dual tropical and extratropical personality, and the enormous spread of its high wind. We examine Sandy's weather impacts, which include not only drenching rain and extremely heavy snowfall, but an astronomically enhanced storm tide, Great Lakes coastal flooding, and an unusual flip-flop in surface temperature extremes. The advance prediction of Sandy's track and intensity was excellent, but we discuss why the future prediction of major storms may suffer. Finally, given the "hot button" topic of Sandy and global warming, we place Sandy in the context of devastating storms that played out long before global temperature began its upward surge in the 1970s.

Superstorm Sandy: A Day-by-Day Chronology

We illustrate the complex, 10-day life history of Sandy in Figure 2. The left side shows the track

of the storm (black dashed line), the width of the hurricane-force sustained winds (red/blue swath overlaid on the track), and the width of the tropical-storm force winds (yellow swath). Dots along the track are colored according to maximum sustained winds. The middle timeline shows how Sandy's vortex categorization changed. Red colors indicate a purely tropical vortex with central warm core, blue colors indicate a cold-core, extratropical vortex, and intermediate shades imply a hybrid system. Finally, the right side presents color-enhanced satellite snapshots of the storm during its many phases.

The Tropical Phase

Sandy's tropical phase spanned the storm's first five days. The storm incubated from a tropical easterly wave off Africa. On October 22, the wave disturbance was declared a tropical depression in the southern Caribbean. When thunderstorms coalesced into curved bands, Tropical Storm Sandy was born as vortex winds tipped 39 mph. Over the next day, the storm moved very little but continued to strengthen. Sandy was embedded in an environment favorable for rapid deepening: Very warm ocean water, weak wind shear, and deep tropical moisture.

On October 23, winds increased to 60 mph. Sandy was now moving north toward Jamaica and developed a ragged eye. Bursts of intense thunderstorms fired continuously over the storm center. An upper-level trough of low pressure southwest of Cuba enhanced the storm's outflow. Air drawn out of the storm's top allowed pressure to drop more rapidly at the surface.

At 11 a.m. on October 24, Sandy was declared a hurricane, as its winds reached 74 mph. Early that morning, a violent eruption of thunderstorms pushed cloud tops into extremely frigid upper layers of the atmosphere, where the cloud top temperature fell to -104°F . Thunderclouds of this depth and vigor are rarely observed. At 3 p.m., Sandy crossed Jamaica with 80-mph winds. In spite of this, pressure rapidly declined, and winds surged to 90 mph.

This was the worst possible situation for Cuba, which lay next in the storm's path. Landfall occurred early in the morning of October 25, near Santiago. Storm winds rocketed up to 105 mph. Sandy was now a strong Category 2 hurricane, bordering on Category 3 status. But as the vortex interacted with Cuba's steep cordillera, the eye disappeared. Sandy emerged from the north coast of Cuba as a 90-mph, Category 1 storm.

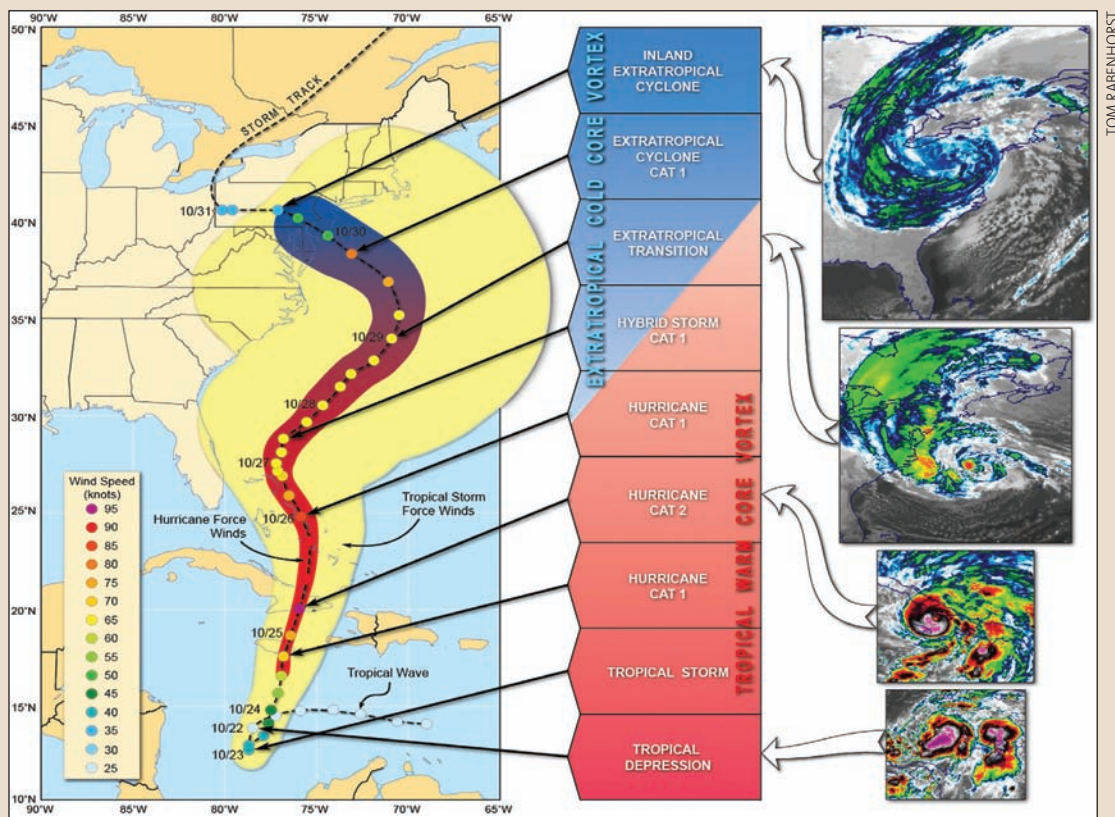


Figure 2. The 10-day life history of Sandy. The left side shows the track of the storm (black dashed line), width of hurricane-force sustained winds (red-blue swath overlaid on the track), and width of tropical-storm force winds (yellow swath). The middle timeline shows how Sandy's vortex categorization changed. Red colors indicate a purely tropical vortex with central warm core, blue colors indicate a cold-core, extratropical vortex, and intermediate shades imply a hybrid system. The right side presents color-enhanced satellite snapshots of the storm during its many phases.



Flooding and damage along Albany Avenue in Atlantic City.

Transition to a Hybrid Storm

Sandy's transit of Cuba took a toll, but forecasters could not breathe easy. For several days, the medium-range prediction models suggested that once Sandy moved north of Cuba, the tropical vortex would encounter a new energy source—a trough of low pressure originating in mid-latitudes. The trough would ventilate the storm, and cold air in its core—when juxtaposed against Sandy's warm tropical eye—would create a temperature contrast (gradient) in upper levels. Such gradients sustain large cyclonic vortices in mid-latitudes. The models suggested that the purely tropical Sandy would morph into a hybrid-type storm—one that draws energy from both the warm tropical waters and the circulation pattern in the middle and upper atmosphere. The infusion of new energy would sustain Sandy against the adverse effects of increasing wind shear. Additionally, all models were predicting a significant expansion of the vortex.

As Figure 1 shows, the footprint of winds swelled on October 26. Strong winds in the trough pushed thunderstorms away from the low-level circulation center. On October 27, cooler air from the North American continent began to wrap toward the storm center from south and west. As this cool air approached the still-warm

tropical core, Sandy began to transform into an extratropical cyclone—one that is fueled mainly by strong temperature contrasts. Sandy was now a fully hybrid storm, holding winds at a steady 70-75 mph.

On October 28, Sandy still maintained a hurricane-like inner core, and tropical storm-force winds ballooned outward by a factor of four, making the storm the size of a very large Nor'easter. Forecasters were now gravely concerned for the East Coast, not just because of the huge wind field, but also because most of the forecast models turned Sandy straight into the mid-Atlantic. This exceptionally rare "left hook" was being driven by two factors: (1) a strong trough of low pressure over the Carolinas, with counterclockwise winds aloft, and (2) a blocking ridge of high pressure over Newfoundland, with clockwise rotating winds. Sandy was like a cog stuck between two very large gears, acting to pull the storm inexorably inland.

Extratropical Transition

On October 29-30, two significant changes took place. First, the storm was in the throes of extratropical transition. Cold air invaded the core from the west and south, while warm tropical air advanced westward. This collision created a sys-

tem of weather fronts. The process was enhanced by the arrival of a strong cold front along the East Coast, which became assimilated into the vortex. Second, Sandy moved over the Atlantic's Gulf Stream—a narrow, warm current coursing northward along the Eastern Seaboard from Florida. Forecasters feared that the Gulf Stream would impel additional energy into Sandy's circulation. Indeed, tropical-like thunderstorms flared within the center, and a new eye developed on October 30. Storm winds increased to 90 mph, hours before landfall, but cooler shelf waters off the Jersey shore arrested further intensification.

Inland Extratropical Cyclone

Superstorm Sandy, now declared a fully extratropical system, slammed into Atlantic City as a strong Category 1 (90 mph) at 8 p.m. on October 30. Once inland, the winds began to rapidly decline, down to 75 mph at 11 p.m. The storm center moved due westward, north of and parallel to the Mason-Dixon line. The next day, Sandy loitered in the vicinity of Pittsburgh, Pennsylvania, then executed a tight right turn over Lake Erie, accelerating toward Ontario. The satellite image in the upper right corner of Figure 1 shows the tightly coiled swirl of an enormous, inland vortex, bearing no resemblance to its comparatively tiny progenitor that had formed south of Cuba six days earlier.

New Scientific Questions Raised by Superstorm Sandy

Figure 3 presents a summary of Sandy's key structural differences as a purely tropical system, as well as its extratropical phase. We show the aerial coverage and intensity of surface winds (colors), weather fronts, the storm track, and upper-level flow features. These include the strong outflow of winds (white arrows) north of Sandy as a tropical cyclone, and the merger of Sandy with an intense upper-level trough as it approached the East Coast.

The exceptional structure of Sandy raises some new questions about the behavior of hybrid-type storms. The manner in which Sandy switched energy sources, acquiring dual tropical and extratropical sources for some time, requires investigation. How did the complex thermal structure of the inner core, with interacting warm and cold regions, evolve? Why did the wind field become so enormous? Some of the expansion resulted from Sandy's weakening north of Cuba, but the storm's interaction with a strong mid-latitude trough likely played an important role, albeit one that is not well understood.

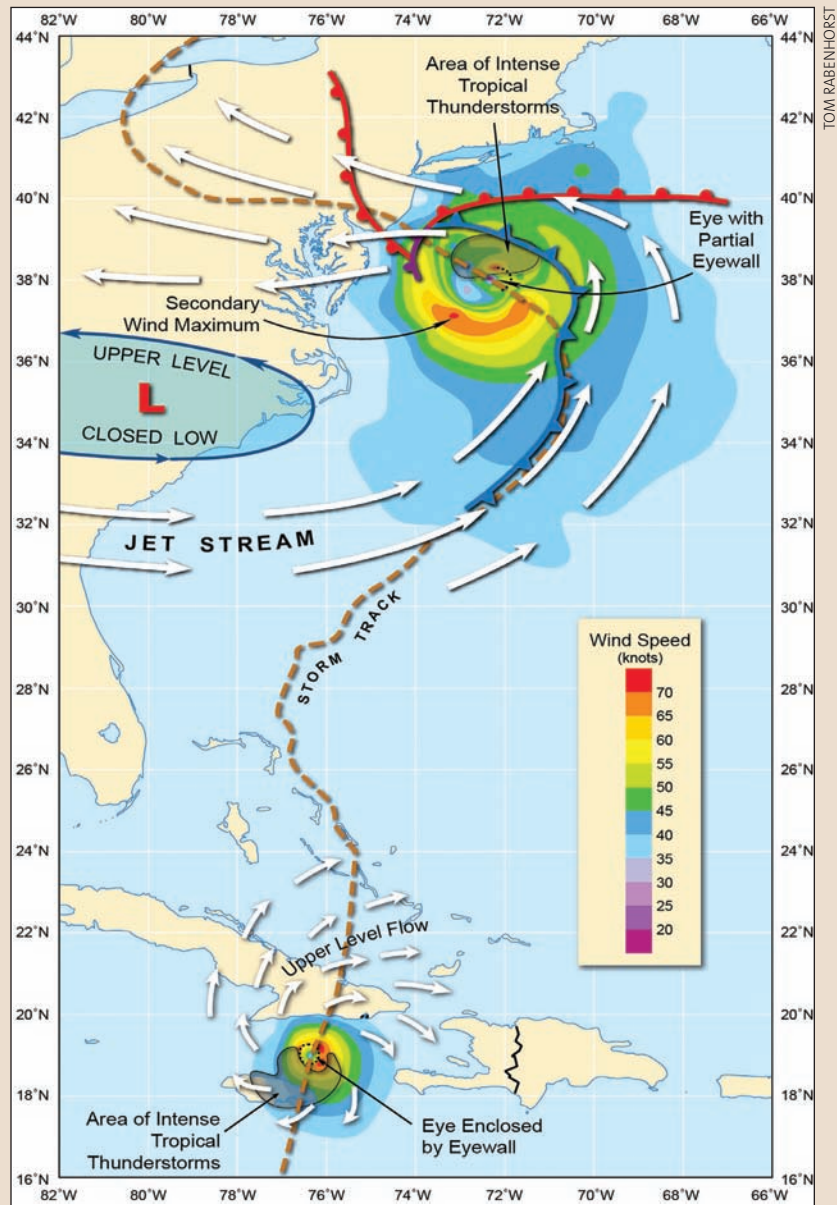


Figure 3. A summary of Sandy's key structural differences as a purely tropical system, and its extratropical phase. We show the areal coverage and intensity of surface winds (colors), weather fronts, the storm track, and upper-level flow features. These include the strong outflow of winds (white arrows) north of Sandy as a tropical cyclone, and merger of Sandy with an intense trough while approaching the East Coast.

Another aspect shown in Figure 3 is the marked asymmetry of strong winds around the vortex center. In purely tropical cyclones, stronger winds develop to the right of the track, where the direction of the swirling wind adds to the movement of the storm. As Sandy approached Cuba, the strongest winds (81 mph; red colors) lay to the right of track. But hybrid and extratropical Sandy developed a second wind maximum to the left of track, at very large radius (100-150 miles) from the storm center. The broad wind max to the south of Sandy (orange swath) was not associated with the storm's tropical eyewall. This

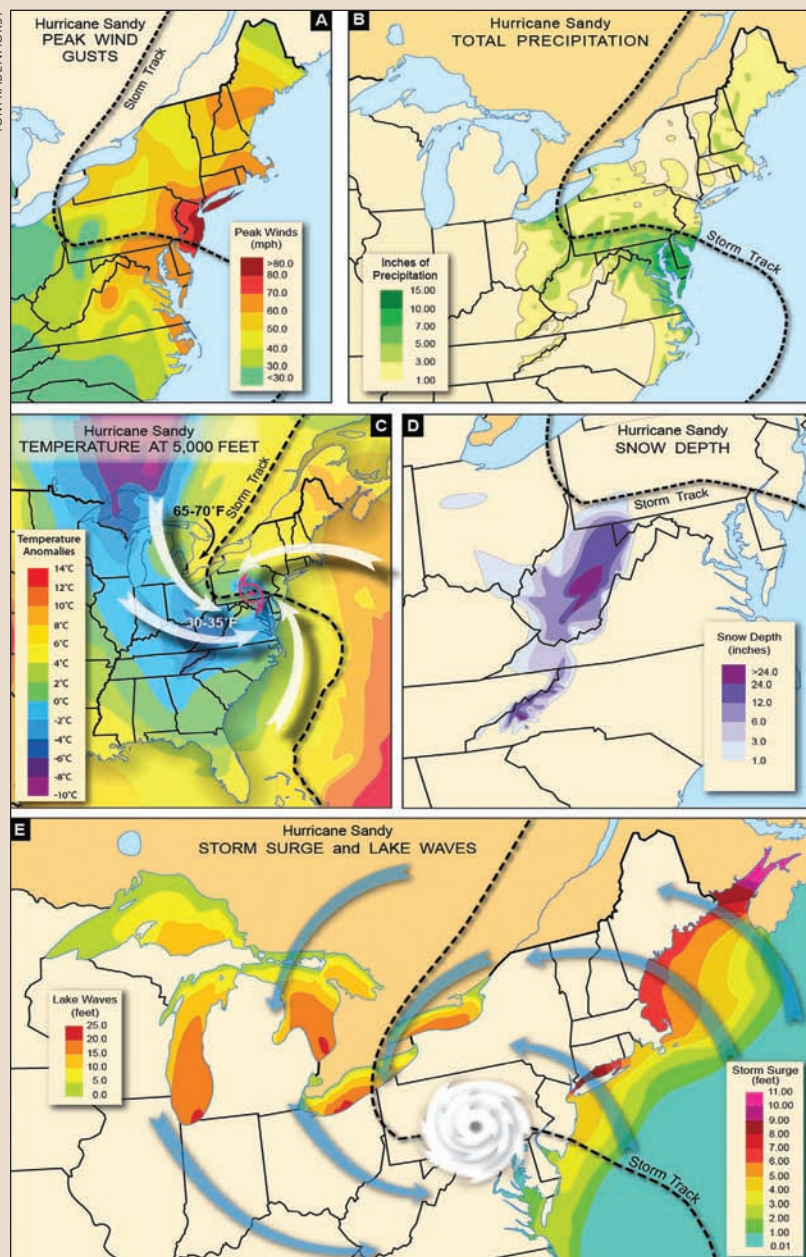


Figure 4. The five principal impacts of Hurricane Sandy.

is a very unusual type of wind pattern—clearly related to Sandy’s hybrid structure—but the origins of the wind’s kinetic energy remain somewhat speculative.

Sandy’s Disparate Impacts: Five Categories of Damage

Superstorm Sandy revealed the great many forms of destructive weather that a large, intense cyclonic vortex is capable of generating. **More than 65 million inhabitants along the Eastern Seaboard felt significant effects.** The type and intensity of weather impact depended not only on geographic location, but also distance from the storm center and side of track. We have graphi-

cally summarized the five principal impacts in Figure 4.

High wind provided the single most widely felt impact, and it extended from the East Coast to the Great Lakes. Peak gusts are mapped in Panel A. **The highest gusts, 90-100 mph, occurred along the Jersey shore and over New York City.** This region was not only closest to the storm center at landfall, but also to the right of the track. Sandy accelerated as it approached the Jersey shore, increasing the wind asymmetry across the storm’s core (wind blowing *from* the east adds to the storm motion). Furthermore, air flowing from the west, left of track (offshore flow), was slowed considerably by surface friction. Note the enhancement of winds along the elevated terrain of the central Appalachians; in tropical cyclones, storm winds are usually 20-30 percent faster just a few thousand feet above mean sea level.

Closely related to the wind field is the height of storm surge along the coast, and raised waves on the Great Lakes (Panel E). Peak storm surge—the mound of seawater that is pushed inland by the wind—occurred to the right of the track, where sustained onshore flow was strongest. The surge was further concentrated within various coastal inlets and bights. Highest surge values were 9-10 feet. **However, storm surge must be distinguished from storm tide.** During Sandy, there were two unusually large, astronomical high tides (so-called spring tides) that resulted from the full moon phase. Storm tide approached 14 feet in the NYC region. **Northwesterly winds gusting to 65 mph over the Great Lakes (more than 600 miles from storm center!) raised waves of up to 23 feet.** These waves pounded the Lakes’ lee shores along Gary, Indiana, and Cleveland, Ohio.

Panels B and D show the distribution of precipitation that fell around the vortex. **Heavy rain is to be expected during landfall of a hurricane, and Sandy proved no exception, with widespread amounts exceeding five inches.** A swath of 10-13 inches fell left of the track. But the biggest surprise from Sandy was heavy, early season snowfall, when tropical moisture combined with subfreezing air in the core of the upper-level trough. The bullseye of accumulating snow was over Central West Virginia. Three feet accumulated in Richwood, West Virginia, and Wolf Laurel, North Carolina. The high elevation of the ground played a role in dropping low-level temperatures to freezing. Also, the high liquid water content of the tropical air mass, combined with temperatures hovering near 32°F, produced a very wet and heavy snow. For many hours, whiteout conditions prevailed as large, sticky flakes were blown sideways in high wind.



Snow from Hurricane Sandy in West Virginia.

Daily maximum temperature across much of the East experienced unusual swings. Extratropical cyclones such as Sandy are, in essence, giant mixers that pull in cold air from the northwest and warm air from the southeast. But Sandy moved inland in a highly occluded, or tightly coiled, configuration. This means that the coldest inland air actually approached the system from the southwest, while warm oceanic air arrived from the northeast. On October 30, while temperatures barely climbed above freezing (32°F) hundreds of miles to the south of Sandy, high temperatures climbed to nearly 70°F in Montreal, Quebec, Canada.

In all, 24 states experienced direct impacts from Superstorm Sandy. These included cancellation of 20,000 airline flights from October 27-November 1, 8.6 million power outages in 17 states (some lasting for weeks), and unprecedented disruption of rail networks across the Northeast. Superstorm Sandy is often compared to Katrina in terms of impacts. In terms of lives

lost, approximately 1800 perished in Katrina, and only 125 died as a result of Sandy. But because of the high population density in the Northeast, Sandy had a much greater impact in terms of destruction of homes totally destroyed or heavily damaged and businesses: the total number of buildings destroyed is 233,000 for Katrina and 570,000 for Sandy.

Predicting Superstorm Sandy and Some New Concerns

The earliest predictions of a significant East Coast impact were made 8-9 days prior to landfall; these highlight the improved success of medium-range (7-10 day) forecast models. The National Hurricane Center narrowed down landfall to the New Jersey shore 4-5 days in advance. Intensity prediction, which is less skillful than track prediction, fared better than average for Sandy.

Will future prediction of storms enjoy similar skill? One likely obstacle is our aging weather satellite surveillance system. Some components,

such as the polar orbiting satellites, are nearing the end of their expected lifespans. The replacements won't be launched until 2017 and 2023. Meteorologists warn of a significant gap in satellite coverage, lasting at least one or two years. Satellites provide crucial information on hurricanes and Nor'easters while they are over the oceans, including storm intensity, size, location, and movement. They also feed data to forecast models—the very medium-range models that performed so well during Sandy. A research group at the European Center for Medium-Range Weather Forecasting (ECMWF) recently revealed what a gap in satellite coverage would do. The ECMWF, which had created one of the most skillful predictions of Sandy, re-ran its simulations of the storm while withholding the polar satellite data. For the prediction made five days prior to landfall, Superstorm Sandy was located hundreds of miles offshore—a clear miss for the mid-Atlantic on that day.

Not surprisingly, Superstorm Sandy provoked much discussion about the role of global warming

in the formation of superstorms. The meteorological community has emphasized that no superstorm, such as Sandy, is directly triggered by the warming environment. However, there is concern that the rise in ocean temperature imparts an incremental increase in intensity, and a rise in atmospheric water vapor slightly enhances heavy rainfall. But we must consider another potential impact on a storm such as Sandy, which derived a large fraction of its total energy from the mid-latitude jet stream. In a warmer world, the high northern latitudes experience the greatest warming. Jet streams are sustained by the temperature gradient between the poles and the tropics. As this gradient relaxes, one would expect the jet stream to weaken. Perhaps this effect diminished Sandy, in its hybrid and extratropical phases, by an incremental amount.

Prior to the warm-up that began in the 1970s, there have been other storms worthy of “superstorm” status. In fact, many meteorologists regard the 1950 Great Appalachian Storm as the 20th century's event with the farthest-reaching



Flooded Avenue C at East 6th Street in Manhattan's East Village neighborhood of LoIsaida, moments before the Con Edison power substation on 14th Street and Avenue C blew up.



FEMA/WALT JENNINGS

Rising floodwaters in the aftermath of Hurricane Sandy caused water damage in hundreds of businesses in Bergen County, New Jersey. Businesses lost millions of dollars in inventory goods when they were soaked with floodwaters.

impacts. The storm occurred during a multidecade period of global cooling. While the storm did not begin its life as a tropical cyclone, it was a late-November Nor'easter that, like Sandy, was forced due west, against the prevailing steering current, by a blocking ridge of high pressure near Greenland. On the warm side of the storm, hurricane-force winds, coastal flooding (including inundation of Manhattan), and flooding rains lashed the East Coast, while heavy snow, up to 57 inches, fell across Appalachia and the Ohio Valley.

Before rushing to “judge a storm by its climate cover,” however, it’s useful to consider the lesson of these two superstorms, which were separated by more than 60 years and occurred within very different climate trending regimes. **W**

JEFFREY B. HALVERSON is an Associate Professor in the University of Maryland Baltimore County’s (UMBC) Department of Geography and Environmental Systems. THOMAS RABENHORST is a Senior Lecturer and Director of Cartographic Services at UMBC’s Department of Geography and Environmental Systems. They are currently authoring a textbook on severe storms and hazardous weather, to be published by Oxford University Press.



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Flooding in Marblehead, Massachusetts, caused by Hurricane Sandy.